Group Number: 31

Predicting Cricket Pitch Conditions using Machine Learning for Data-Driven Match Strategy

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***Abstract*—Pitch conditions strongly influence cricket match outcomes, yet strategic decisions such as the toss call and the balance of pace and spin in the playing XI continue to rely largely on visual inspection and venue-specific experience. This reliance is further limited by the absence of public datasets that quantify pitch behaviour, and it becomes unreliable at unfamiliar venues. This paper proposes a machine learning framework that infers pitch behaviour from openly available data. Using ball-by-ball records from Cricsheet, a Pitch Behaviour Profile is derived for each venue from run rate, boundary percentage, dismissal rate, and the distribution of wickets between pace and spin bowling. A Pitch Wear Index is further introduced to quantify surface deterioration by measuring the change in scoring and wicket patterns between the first and second innings, normalised for team strength. These features are combined with historical weather attributes, namely temperature, humidity, rainfall, and wind speed, obtained from the Open-Meteo archive. Random Forest and Gradient Boosting classifiers are trained to categorise pitches as batting-friendly, balanced, pace-friendly, or spin-friendly, and are evaluated using accuracy, precision, recall, and F1-score under time-aware cross-validation. To address venues with limited or no historical records, a similarity-based cold-start approach matches an unfamiliar ground to comparable venues using their behaviour profiles and climatic conditions. SHAP-based explanations identify the factors driving each prediction, and the outputs are translated into bat-first or field-first toss recommendations. Relying exclusively on free public data, the proposed framework is reproducible and offers an objective, interpretable basis for match strategy.**

***Index Terms*—cricket analytics, pitch behaviour classification, pitch wear index, cold-start prediction, explainable machine learning**